316 Lab

Problem: Sum of Digits Using Recursion

Time: 30 minutes	Marks: 10

Problem Description

Write an 8086 Assembly Language program that reads a non-negative decimal integer and calculates the sum of all its decimal digits. The digit-sum calculation must be performed using a recursive procedure.

For example, if the input is **572**, its digits are 5, 7, and 2. Therefore, the result is **5 + 7 + 2 = 14**.

Your recursive procedure should process one digit at a time and continue calling itself until no digits remain. The program must correctly handle both single-digit and multi-digit numbers, including the input 0.

Input Specification

- The input contains one non-negative decimal integer N.
- Assume that 0 <= N <= 65535.
- The input must be read from the keyboard.

Output Specification

- Print the sum of the decimal digits of N.
- The result must be displayed in decimal form.

Sample Input and Output

Sample	Input	Output
1	572	14
2	905	14
3	7	7
4	0	0
5	65535	24

Submission Note

Submit the complete 8086 assembly source code. The recursive procedure should be clearly identifiable, and brief comments should indicate the base case and the recursive call.